

The Environmental Kuznets Curve at the State Level in India: Economic Growth and Air Pollution

Project Proposal

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1. Introduction and Motivation

Air pollution is one of India's most pressing public and environmental problems, with several Indian cities regularly recording among the world's highest particulate concentrations. A long-standing question in environmental economics is whether pollution and economic growth are linked by an Environmental Kuznets Curve (EKC), the hypothesis that pollution rises in the early stages of economic development but eventually falls once a region becomes wealthy enough to prioritise environmental quality (Grossman and Krueger, 1991). This project investigates whether that relationship holds at the state level in India, using satellite-derived pollution data and official state economic accounts.

My interest in this question is also personal. Having grown up in India, air pollution was not an abstract statistic but part of ordinary life: seasonal smog that turned daylight hazy, days when going outside meant a sore throat, and the familiar public argument that pollution is simply the price a developing country pays for growth. That last claim is, in effect, the Environmental Kuznets Curve stated informally, and it is one I have heard repeated far more often than I have seen tested. Whether it actually holds for Indian states, and if so at what income level the relationship turns, is therefore a question I have a genuine stake in, and one that statistical analysis can address directly rather than rhetorically.

2. Research Questions

RQ1. Does an inverted-U-shaped Environmental Kuznets Curve relationship hold between economic growth (GSDP per capita) and air pollution (PM_{2.5}) across Indian states?

RQ2. Given genuinely mixed findings in the existing literature on India's EKC, where does the evidence point using state-level data, and does the National Clean Air Programme (2019) correspond to a measurable shift in the pollution-growth relationship?

3. Related Work

The EKC hypothesis originates with Grossman and Krueger (1991), who observed an inverted-U relationship between per-capita income and certain pollutants. Evidence for India remains contested. Farooq, Bhanja, Rather and Dar (2024) find an inverted-N-shaped curve

using ecological footprint data, implying a more complex relationship than a simple inverted-U and that India remains in an increasing-degradation phase. Al-Mulali, Ridwan, Aslan and Raboshuk (2025) find that specific policy interventions, such as electric vehicle incentives and renewable energy programmes, shift the growth-pollution relationship rather than the curve following income alone.

Critically, Wagner (2008, 2015) shows that much of the EKC literature is econometrically unsound: powers of integrated processes are not themselves integrated processes, so applying standard unit-root and cointegration techniques to a specification containing both income and income squared is invalid. Applying appropriate methods substantially weakens the reported evidence for an EKC. This project treats Wagner's critique as a binding methodological constraint rather than a footnote, and its specification is chosen accordingly (Section 5).

4. Data

Air pollution (dependent variable)

The primary source is satellite-derived annual mean surface PM_{2.5} from the Atmospheric Composition Analysis Group at Washington University in St. Louis (van Donkelaar et al., 2021; V5.GL series, approximately 0.01 degree resolution, 1998 onward), distributed free of charge via ACAG and NASA SEDAC. This source is chosen deliberately over ground-monitor data for two substantive reasons. First, India's ground network (CPCB/NAMP) has expanded substantially over time and is heavily concentrated in metropolitan areas, with Delhi and the National Capital Region accounting for a large share of real-time stations while much of rural India remains unmonitored; a state's measured pollution can therefore change simply because monitors were added, confounding measurement change with genuine pollution change. Satellite estimates provide spatially complete, temporally consistent coverage for every state in every year. Second, the longer satellite record extends the panel to roughly twenty-five annual observations per state rather than about ten, materially improving the statistical power of panel tests.

As a validation source, CPCB ground-station data (via OpenAQ's free API and CPCB's own bulletin archive) will be used to check that satellite estimates track ground measurements reasonably in states with good monitor coverage. This serves as a robustness check rather than the primary series.

Economic data (independent variable)

State-wise Gross State Domestic Product is taken from the Ministry of Statistics and Programme Implementation and the Reserve Bank of India's Handbook of Statistics on Indian States, deflated to constant prices and divided by state population to give GSDP per capita. Population figures derive from the 2001 and 2011 Censuses with official inter-censal projections thereafter; see Section 7.

All sources are free and publicly accessible. The full workflow, covering data acquisition, processing of gridded satellite files to state-level aggregates, and estimation, runs on a personal laptop in Python or R, and requires no fieldwork, paid data licences, or specialised hardware.

Data availability

Satellite PM_{2.5} estimates (V5.GL series, annual, gridded): Atmospheric Composition Analysis Group, Washington University in St. Louis, <https://sites.wustl.edu/acag/surface-pm2-5/> (also distributed via NASA SEDAC and <https://www.satpm.org>).

Ground-station air quality, used for validation: OpenAQ, <https://openaq.org> (API documentation at <https://docs.openaq.org>); Central Pollution Control Board daily AQI bulletins, https://cpcb.nic.in/AQI_Bulletin.php; and the CPCB real-time feed hosted on the Government of India open data portal, <https://www.data.gov.in>.

State economic and demographic data: Reserve Bank of India, *Handbook of Statistics on Indian States* and Database on Indian Economy, <https://data.rbi.org.in>; Ministry of Statistics and Programme Implementation, <https://mospi.gov.in>; and Census of India, <https://censusindia.gov.in>.

5. Proposed Methodology

Step 1. Panel construction

Gridded satellite PM_{2.5} is aggregated to state-year means, population-weighted so that estimates reflect human exposure rather than unweighted land area, and merged with real GSDP per capita to form a balanced state-year panel.

Step 2. Diagnostics

Cross-sectional dependence is tested using the Pesaran CD test, since Indian states are spatially and economically interlinked and dependence is expected. Because first-generation panel unit-root tests (Levin, Lin and Chu, 2002; Im, Pesaran and Shin, 2003) assume cross-sectional independence, second-generation tests (Pesaran, 2007, CIPS) are used to assess the integration properties of the underlying series.

Step 3. EKC specification, respecting Wagner's critique

The baseline specification regresses PM_{2.5} on GSDP per capita and its square within a fixed-effects panel model with state and year fixed effects, using Driscoll-Kraay standard errors that are robust to cross-sectional and serial correlation. Following Wagner (2008, 2015), no cointegration claim is made for the quadratic specification, since powers of integrated regressors are not themselves integrated. Results are therefore interpreted as conditional associations within states over time, not as a cointegrating long-run relationship. A Hausman (1978) test confirms the choice of fixed over random effects. If time permits, cointegrating polynomial regression methods (Wagner, 2015) will be explored as an extension.

Step 4. Policy intervention and shock controls

A post-2019 indicator tests whether the National Clean Air Programme corresponds to a shift in the pollution-growth relationship. A separate 2020–21 indicator absorbs the COVID-19 shock, during which output and pollution fell together for reasons unrelated to the EKC mechanism; leaving this uncontrolled in a relatively short panel would materially bias the estimated curve.

Step 5. Interpretation

If a significant inverted-U is found, the income turning point is estimated with confidence intervals, and checked against whether it lies within the observed income range. Turning points

estimated by extrapolation beyond the sample are a known weakness of this literature and will be reported as such rather than presented as forecasts.

6. Significance

This project contributes a state-level analysis of an unresolved question in environmental economics, using consistent satellite pollution data and official Indian economic accounts, and explicitly addressing an econometric critique that much of the applied EKC literature overlooks. It has direct policy relevance: whether growth alone will improve air quality, or whether targeted intervention is required, matters for how India's states approach environmental regulation.

7. Limitations

- Causal identification. Pollution can itself reduce economic output through health and productivity effects, so the income-pollution relationship is jointly determined. This project does not employ instrumental-variable or other causal-identification strategies; results are associational and will be described as such.
- Cross-state spillovers. Pollution crosses state boundaries, and Delhi's air quality in particular is strongly affected by crop-residue burning in Punjab and Haryana. State-level estimates may therefore misattribute pollution to local economic activity. Spatial econometric extensions are noted as future work.
- Population denominator. India's 2021 Census was postponed, with enumeration now scheduled for 2027, so post-2011 per-capita figures rely on projections rather than enumerated counts, introducing measurement error into the key regressor.
- Series consistency. India revised its GSDP base year to the 2011–12 series, and Jammu and Kashmir was reorganised into two union territories in 2019. Both require explicit, documented handling to avoid artificial discontinuities in the panel.
- Satellite estimate error. Satellite-derived PM_{2.5} is modelled rather than directly measured and carries its own uncertainty. The ground-station validation step described in Section 4 partially addresses but does not eliminate this.

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